TIPA Korea Technology and Information Promotion Agency for SMEs

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

Proposer	Name	Dr. Mateusz Balcerzak	Email	mateusz.balcerzak@ifam-dd.fraunhofer.de
Institute Name	Fraunhofer IFAM Dresden			
Address	Winterbergstraße 28, 01277 Dresden, Germany			

2. Technology Overview

Title of Proposed Technology	*Energy storage system for autonomous marine application*			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine (√), Hydrogen (), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Mechanical・Materials		*Energy & Resources*	
	Medium Category	Medium Category	Small Category	Small Category
	Shipbuilding/ Marine Systems	*Hull/Deck and Propulsion System Components*	*New & Renewable Energy*	*Hydrogen*
Concept of Proposed Technology	*Core concept: A modular energy system uses metal hydrides for safe, high-density, low pressure hydrogen storage to power fuel cells on autonomous maritime platforms.* *Main functions: low-pressure storage (for example: 5–30 bar) and thermally or pressure regulated H_2 release via integrated heat exchangers and waste-heat coupling; precise state estimation (pressure/temperature); docking interface for simple refueling; hydride modules with adaptive heat routing; real-time, data-driven range and maintenance models using state-of-the-art sensor technology (e.g. EP3045910B1) (1).* *Example use cases: uncrewed surface marine vehicles, autonomous underwater vehicles, buoys, and subsea observation nodes, offshore platforms.* *Target: TRL-6 demonstrator within 36 months.*			

Rationale for Joint R&D Project	*Autonomous maritime systems require safe, long-duration energy. Metal-hydride hydrogen systems are an alternative to batteries and high-pressure/cryogenic H_2 storage, enabling extended endurance and complementing the deployability of underwater platforms. However, they demand co-development of materials, thermal management, fuel-cell integration, and maritime qualification. Collaboration is essential to integrate these disciplines and meet certification. Fraunhofer IFAM brings nearly 20 years of hydrogen experience, with unique know-how and IP in metal-hydride-based separation, storage, and compression (1-3). Korean SME industry partners contribute use-case definition, platform integration, manufacturability, and supply-chain localization. Joint R&D de-risks the technology and accelerates time-to-TRL-6.*
Key Words	1. *Metal hydride hydrogen storage* 2. *Low-pressure operation (5–30 bar)* 3. *Autonomous maritime platforms* 4. *Modular energy system* 5. *Fuel cell integration*

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (4)	End of the Project	Level (6)
R&D Objectives	*The final objectives will depend on the project scope (strongly depending if the project should aim to develop the hydrogen storage unit for marine application or will target integrated storage unit on the board of maritime vessel. Exemplary objective include:* *1. marine grade metal hydride storage modules operating at 5–30 bar (other pressure ranges also possible if needed) with segmented beds and integrated thermal interfaces for efficient heat flow;* *2. thermal management that couples fuel cell waste heat to controllable hydrogen desorption, enabling reliable start up and operation across seasonal seawater temperatures;* *3. a docking/refueling interface supporting safe refueling under stationary (ports, buoys, fixed platforms) and mobile conditions (service vessels, unmanned ships);* *4. onboard sensing and data logging (pressure, temperature, flow) with standardized datasets and APIs to support real-time estimation of the predictive range;* *5. modular packaging for rapid integration on autonomous surface and underwater platforms.*			
R&D Target Specifications	• *A modular TRL-6 prototype optimized for maximum system-level volumetric energy density, featuring integrated fill-level sensing in metal-hydride modules.* • *Stable, controllable desorption at (for example) 5–30 bar with precise flow control adaptable to application demand.* • *Validated thermal management enabling charge/discharge across seawater temperatures (~2–35 °C).*			
R&D Plan	*Research approach* • *Iterative development cycles across materials, module design, thermal management, refueling, sensing, and system integration. Fraunhofer IFAM leads hydride and system design; Korean small and medium-sized enterprises (SMEs) lead platform integration, manufacturability, and software use cases.* *Overall approach and timeline (36 months total)* *The project is structured into two sequential phases, progressing from concept definition to platform integration and sea tests. The phase one will end with reaching M1 after 6 months of the project.* *Key tasks*			

WP	*Scope*	*Key outputs*	*Milestone alignment*
1	*Requirements specification (Lastenheft), system architecture, Process and Instrumentation Diagram (P&ID), CAD-level digital twin, Bill of Materials (BOM)*	*Approved Lastenheft, P&ID, initial digital twin, released BOM*	*M1*
2	*Thermal management for 2–25 °C seawater*	*Demonstrated charge/discharge control across 2–25 °C*	*M2*
3	*Docking/refueling interface for stationary and mobile operations*	*Validated refueling procedures, tooling, safety checks*	*M2*

4	*Onboard sensing, data logging, standardized APIs (Application Programming Interfaces)*	*Released data interfaces supporting control models from the SME*	*M2*
5	*System build and bench testing (storage, fuel cell, power coupling)*	*Bench-validated prototype subsystems*	*M2*
6	*Platform integration and sea tests*	*Integrated prototype, completed sea tests, TRL-6 demonstrator, as-built digital twin*	*M3*

Some of the key methodologies

- *Arc remelting*
- *pressure–composition–temperature measurements;*
- *Finite Element Analysis of the heat transfer management (FEA);*
- *Engineering;*
- *Construction;*
- *sea water testing.*

Milestones

Year	Timing	Milestone
1	M6	Requirements specification (Lastenheft) and safety plan complete; Process and Instrumentation Diagram (P&ID) completed; basic concept of a digital twin at P&ID level in CAD; lab module validated with controlled desorption at 5–30 bar; Bill of Materials (BOM) released.
2	M24	System-level prototype built and bench-tested (metal-hydride storage, fuel cell, and power-coupling subsystem); operation validated across 2–25 °C seawater conditions; stationary and mobile refueling validated; standardized data interfaces released.
3	M36	Prototype (metal-hydride storage, fuel cell, and power-coupling subsystem) integrated into the platform defined in the requirements specification; sea tests completed; Technology Readiness Level (TRL)-6 demonstrator achieved; as-built digital twin delivered for scale-up.

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	*Fraunhofer IFAM will contribute/responsible to/for WP1–WP4.* • *WP1-2: Material selection and characterization of metal hydride storage; conceptual pressure vessel design (CAD/FEA models, segmented beds, integrated thermal interfaces); thermal management models including waste-heat coupling. Deliverables: PCT/material data packages, design concepts, thermal modeling results.* • *WP3: Specifications for docking/refueling interfaces, operating limits, and safety procedures. Deliverables: interface specifications and process descriptions for stationary/mobile refueling.* • *WP4: Sensor specifications, data models, and algorithm concept for state and range estimation. Deliverables: specification documents, data models, algorithm descriptions.* *Resources: materials expertise, H2 material characterization (PCT), system and thermal modeling, FEA/CAD, existing IP; and specification documents.*
Expected R&R of Korean SME Partner	*Korean SME contribute/responsible to/for WP1, WP5 and WP6,* • *WP1: Lead requirements and Korean compliance; develop safety plan, P&ID, CAD-level digital twin, and BOM.* • *WP5: Perform all physical work—detailed engineering, procurement, fabrication, assembly, and commissioning; build bench prototype and validate operation across ~2–35 °C seawater, including thermal management, fuel-cell coupling, and refueling hardware.* • *WP6: Integrate on target platforms; conduct sea trials; deliver TRL-6 demonstrator, as-built digital twin, and operating/maintenance procedures.* *Resources: maritime construction expertise, hydrogen-capable test environment (ventilation, gas detection, emergency protocols), workshop capacity (fabrication, pressure testing), and sea access (vessel, permits).* *Collaboration efforts: biweekly design reviews with IFAM, shared data hub for P&ID-/CAD/BOM-files, and joint test plans.*

5. Expected Outcomes/Impact

Expected Outcomes and Impact	• *TRL-6 outcome: integrated, sea-tested prototype of a modular metal-hydride/fuel-cell energy system for autonomous maritime platforms.* • *Technological achievements: marine-grade low-pressure (5–30 bar) hydride modules with integrated thermal management; stable charge/discharge across ~2–35 °C seawater; docking/refueling interfaces; sensing, data models, and range-estimation algorithms; as-built digital twin.* • *Commercialization potential: hydrogen-based power for USVs/AUVs/buoys; scalable manufacturing via Korean SME.* • *Industrial/economic impact: localized supply chain and capabilities in Korea; reduced operational emissions and logistics; enabling longer autonomous missions; spillover to remote/stationary power; export potential and new jobs in hydrogen maritime technologies.*

6. References

(1) EP3045910 B1 patent, Messeinrichtung und Messverfahren zur Bestimmung der Menge eines in einem Speicher aufgenommenen Gases an einem porösen Speichermaterial

(2) EP 3 118 511 B1 patent, Speicherelement für Gase

(3) DE 10 2020 204 214 B4 patent, Behälter zum Speichern von Wasserstoff

☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization